Gene Pulser® Electroprotocols

* We recommend adapting this protocol to use the Gene Pulser electroporation buffer (catalog #165-2676, 165-2677), which increases cell viability and transfection efficiency in mammalian cell lines.

Cell Type Mammalian, adherent, suspension

Species Used Human, HeLa, epithelial carcinoma

Molecules Electroporated DNA: plasmid

Before the Pulse

Cell Growth Medium Not given

Growth Phase at Harvest 50 to 80%

Pre-pulse Incubation Ice, 10 minutes

Wash Solution Not given

The Pulse

Instruments Used Not given

Electroporation Temperature Room temperature

Electroporation Medium* Phosphate Buffered Saline

Cell Density 1 x 10 (7) cells / ml

Volume of Cells 800 µl

DNA Concentration 1 µg / ml

DNA Resuspension Buffer TE Buffer (10 mM Tris, 1 mM EDTA)

Volume of DNA 5 µl

Cuvette Gap 0.4 cm

Voltage 0.25 kV

Field Strength 0.625 kV/cm

Capacitor 960 µF

Resistor (Pulse Controller) none Ω

Time Constant 18 to 22 msec

After the Pulse

Outgrowth Medium Not given

Outgrowth Temperature Not given

Length of Incubation Not given

Selection Method or Assay Used Not given

Electroporation Efficiency Not given

Per Cent Survival Not given

Relevant Publications and/or Comments

Note: exponential values designated in parentheses.
PBS: 1x = 8g NaCl,0.2g KCl,0.2g KH2PO4, 1.15g Na2HPO4

Name of Submittor Phang Seng Meng, Graduate Student

Institution Address Dr. Thomas Leong Laboratory
National University of Singapore
Institute Molecular Cell & Biology
Kent Ridge Crecent
Singapore 0511

Survey Number 095